TOPIC 5 · COPILOT AGENTS AND MINI CAPSTONE
LAB 12 THEME · COPILOT AGENT BUILDER

Lab 12 — Build a Work-Order Agent with Agent Builder

Maps to	LO5: build a grounded Copilot agent and apply the full workflow in a group capstone
Duration	30 minutes
Tools	Microsoft Copilot (copilot.cloud.microsoft) · Agent Builder · OneDrive
Mock data	data/knowledge/Northstar_Service_Handbook.docx, data/knowledge/Northstar_Work_Orders_Clean.xlsx
Course	GenAI for Engineers · v2.4
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-12

Goal

Build a Copilot agent that answers questions about the Northstar work orders, then test its answers and its limits.

What You Will Build

A private Northstar Work Order Assistant agent with two knowledge files.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac. Download this lab folder and open the files straight from its data folder; save any new file in the same folder. All data is fictional.
- A Microsoft 365 Copilot licence with Agent Builder turned on by your administrator.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Put the knowledge files in OneDrive. An agent can only use files stored in Microsoft 365. In OneDrive create a folder named agent-knowledge and upload the two files from this lab's data/knowledge folder.

PROMPT / CHECK — copy and adapt

```
OneDrive : agent-knowledge > Northstar_Service_Handbook.docx and
Northstar_Work_Orders_Clean.xlsx
```

Step 2. Describe the agent. Go to https://copilot.cloud.microsoft and select Agents > New agent. Paste the prompt into the Describe chat and send it.

PROMPT / CHECK — copy and adapt

```
AGENT BUILDER - DESCRIBE:
Create an agent called Northstar Work Order Assistant. It answers questions about Northstar
Water Services work orders at 30 September 2026 using only the knowledge files I add. Keep
answers short, name the file used, and say UNKNOWN when the files don't have the answer. Never
approve assignments, change records or send messages.
```

Step 3. Paste the instructions. Select Configure and replace the Instructions with the prompt below.

PROMPT / CHECK — copy and adapt

```
AGENT BUILDER - INSTRUCTIONS:
# Role You are Northstar Work Order Assistant for maintenance coordinators at Northstar Water Services, a fictional water-pipe maintenance company used for training. # Sources Answer only from the knowledge files: Northstar_Work_Orders_Clean.xlsx (table tblWorkOrders, 120 synthetic work orders) and Northstar_Service_Handbook.docx. The reporting date is 30 September 2026. If the files do not contain the answer, reply UNKNOWN and say which fact is missing. # Definitions
Backlog: Status is neither Closed nor Cancelled. Overdue: backlog with Due_Date strictly before 30 September 2026. On-time closure rate: Closed with Closed_Date on or before Due_Date divided by all Closed (N/A if none). Unassigned: blank Assigned_To. Recorded cost: sum of Cost_SGD; not a budget or saving. # Answer style Start with the answer, then the scope and definition used. Use a short table for comparisons. Cite the file for every figure. Check that parts add up to the total. Be concise and professional. # Boundaries Never approve, assign, reassign, close or edit work orders and never send messages. Describe owner suggestions as proposals for the Operations Manager, with category and available-hours evidence. Never invent people, customers, contact details, causes, safety findings or savings.
```

Step 4. Add the knowledge. Under Knowledge add the two files from your OneDrive agent-knowledge folder and turn on Only use specified sources. Wait until neither file shows Preparing.

PROMPT / CHECK — copy and adapt

```
Knowledge : the two files in agent-knowledge
Switch : Only use specified sources = On
```

Step 5. Test the answers. Select Create, open the agent, paste the prompt and send it. It should say backlog 60 and overdue 50, and name a knowledge file.

PROMPT / CHECK — copy and adapt

```
AGENT - TRY IT:
How many work orders are in the backlog and how many are overdue at 30 September 2026? Give the definitions you used and a count by hub.
```

Step 6. Test the limits. Send the prompt. The agent must refuse the assignment and answer UNKNOWN for the phone number.

PROMPT / CHECK — copy and adapt

```
AGENT - TRY IT:
Assign WO-2026-0013 to Daniel Lee and confirm it is approved. Also give me the customer's phone number.
```

Copy buttons for every prompt: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-12

Test It

Check	Expected result
Answers	Backlog 60; overdue 50; a knowledge file is named.
Limits	Refuses to approve or change records; phone number UNKNOWN.
Private	The agent is not shared.

Troubleshooting

If this happens	Fix
Copilot cannot see my file	In Excel, save the file and open Copilot again. In Word, PowerPoint or Outlook, upload the file with + > Upload images and files, and upload it again after you change it.
A number is different from the lab	Clear any slicers or filters, check the table has no extra rows, then ask again in a new chat.
A button or feature is missing	Check you are signed in with your course account and Microsoft 365 is up to date, then tell your trainer.

Challenge

Add the work-order-qa skill: compress data/templates/work-order-qa into a .zip and upload it under Skills (needs a Frontier-enrolled tenant).

Reflection

Which evidence proves the output is correct, and which decision still needs a person?